



Research paper

Effect of perceived social support, comorbid psychiatric disorders, and pregnancy complications on adaptation to pregnancy in adolescents: A comparison study from southeastern Turkey

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ABSTRACT

Objective: Adolescent pregnancies are categorized as high risk due to their adverse obstetric and neonatal outcomes. The aim of our study was to investigate the well-being of pregnant adolescents by assessing pregnancy complications, depression, anxiety, social support, and adaptation to pregnancy.

Methods: Pregnant adolescents who presented to the Obstetrics and Gynecology Outpatient Clinic of Hatay Training and Research Hospital between July 2022 and January 2023 were included in the study. Pregnant adults were also included as a control group during the same period. All participants were assessed using the Beck Depression Inventory, Multidimensional Scale of Perceived Social Support, Perinatal Anxiety Screening Scale, and Prenatal Self-Evaluation Questionnaire.

Results: A total of 61 pregnant adolescents were enrolled in the study and were compared with 116 adults. Our findings indicate that pregnant adolescents tend to exhibit a higher rate of pregnancy complications in comparison with pregnant adults, with complication rates of 14.8 % and 6.1 %, respectively ($p = 0.057$). The presence of complications had a negative impact on the acceptance of pregnancy in adolescents unlike in adults ($p = 0.022$). The relationship of pregnant adolescents with their partners was negatively affected by pregnancy complications ($p = 0.046$). Perceived social support (especially that of a partner) significantly reduced depression and anxiety scores in pregnant adolescents during the prenatal period ($p = 0.002$).

Conclusion: Adolescent pregnancies have negative psychiatric consequences even in traditional societies where child marriage is accepted. Future studies should consider variables that may reduce the impact of sociocultural factors on adolescent pregnancy.

1. Introduction

The World Health Organization defines adolescent pregnancy as pregnancy occurring in individuals aged between 10 and 19 [1]. Pregnancy in adolescence is considered a significant public health concern worldwide because the mother has not yet reached sufficient physical, psychological, social, and hormonal maturity and because of the risks related to the child being born [2]. Pregnancy and parenthood may negatively affect normative developmental processes in this critical period, while the dual biological transitions of adolescence and

pregnancy can increase individuals' psychological and physical vulnerability [3]. Adolescent pregnancies are classified as high risk, since these factors lead to adverse obstetric and neonatal outcomes [4]. Studies examining the effects of maternal age on perinatal outcomes have found that early maternal age (adolescent) alone increases the possibility of adverse outcomes [5].

The World Health Organization states that pregnancy and childbirth complications are the leading cause of death for girls aged 15–19 worldwide [6]. In addition, adolescents are at high risk for adverse health outcomes, including pregnancy complications [7] and perinatal

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and postnatal depression [8]. Although studies have found a higher risk of depression in pregnant adolescents than in pregnant adults and non-pregnant adolescents [9], little is known about the risk factors for this condition [10]. While some risk factors may appear to be similar to those in adults, evidence suggests that other factors are specific to adolescence. Postnatal depression in adolescence was found to be associated with a lack of adequate social support, whereas adult depression was associated with an unplanned pregnancy, stress, and economic status [11]. Furthermore, additional studies have indicated a negative correlation between social support and perinatal depression in adolescent mothers [12]. In addition, it has been reported that the new mother's attachment pattern, which is of great importance for the quality of her relationship with her mother and partner, determines the process of adaptation to pregnancy and the attachment that she will develop with her baby [13].

In addition to obstetric and neonatal complications, adolescent pregnancies also lead to social issues such as limited access to education (falling behind or dropping out of school), inability to secure employment, and failure to attain a proper social status [14]. Quality of life is a comprehensive concept that is influenced by physical health, psychological well-being, personal beliefs, social relationships, and environmental factors [15]. Pregnancy, a transitional period with significant physical and emotional changes, is reported to adversely affect the quality of life in adolescents [2].

Annually, an estimated 21 million adolescent girls (aged 15–19) in developing countries experience pregnancy, with roughly 12 million giving birth [16]. Factors such as sociocultural, socioeconomic, and sociodemographic characteristics, as well as societal moral judgments, play a role in adolescent pregnancies. In developed societies, adolescent pregnancies often occur outside of marriage, whereas in developing countries, adolescent marriages are more socially accepted. It is known that most adolescent pregnancies in Turkey occur within the context of marriage, with available family support, and are often planned and desired [17]. The literature has not sufficiently investigated the relationship between adolescent pregnancies and their psychological impact. The aim of our study was to investigate the well-being of pregnant adolescents by assessing medical complications, depression, anxiety, social support, and adaptation to pregnancy. The secondary objectives were to describe the population of pregnant adolescents in terms of sociodemographic characteristics and the reasons for their pregnancies and also to compare adolescent pregnancies with adult pregnancies.

2. Methods

This comparative cross-sectional study was conducted after obtaining ethics committee approval (decision no. 2022/07-19) from the Ethics Committee of Hatay Mustafa Kemal University.

2.1. Participant selection

Pregnant adolescent patients were referred to the Child Psychiatry Department because they were minors when they visited the Obstetrics and Gynecology Outpatient Clinic at Hatay Training and Research Hospital. All adolescents who presented between July 2022 and January 2023 were evaluated for inclusion in the study. The Obstetrics and Gynecology Outpatient Clinic reported cases of pregnant, unmarried adolescents under the age of 18 to the relevant authorities for legal action. In Turkey, individuals over the age of 17 can get an official marriage permit upon family request and a court decision. We recruited patients who presented to the Gynecology and Obstetrics Outpatient Clinic at Amasya University and Hatay Training and Research Hospital for inclusion in the group of pregnant adults. We enrolled at least 50 volunteers from each trimester, with the first applicants from each trimester selected. Hatay is a city located in southeastern Turkey, and Hatay Training and Research Hospital is a tertiary hospital that is the second

largest hospital in the region after Hatay Mustafa Kemal University Hospital. Birth rates are very high in the region due to the presence of Syrian refugees, but the study included only Turkish patients. As a result of the post hoc power analysis performed using the G Power program after the study, the power of the study was calculated to be 81 % with the current sample size, taking the alpha error as 0.05 and the effect size as 40 % (calculated according to the current study results).

2.2. Scales and measures

2.2.1. Sociodemographic data form

The form comprised questions about the sociodemographic characteristics of the pregnant women and their families (age, educational level, employment status, total monthly income, maternal educational level, paternal educational level, history of pregnancy loss, presence of living child, and planned pregnancy).

2.2.2. Beck Depression Inventory (BDI)

The scale was developed by Beck et al. [18], and the Turkish validity and reliability study was conducted by Hisli [19]. The purpose of this inventory is to determine the risk for depression and to measure the level and severity of depressive symptoms. Scores are interpreted as 0–9=minimal depression, 10–16=mild depression, 17–29=moderate depression, and 30–63=severe depression; the cut-off score is determined as 17.

2.2.3. Scale of Perceived Social Support (MSPSS)

The Multidimensional Scale of Perceived Social Support (MSPSS), developed by Zimet et al., subjectively evaluates the adequacy of social support received from three different sources: family, friend, special person [20]. The validity and reliability study of the MSPSS Turkish form was conducted by Eker and Arkar [21]. A higher score indicates higher perceived social support.

2.2.4. Anxiety Screening Scale (PASS)

This 31-item scale was developed by Somerville et al. in 2014 [22], and the Turkish validity and reliability study of the scale was conducted in 2018 by Yazıcı et al. [23]. The scores obtained from the scale are evaluated as minimal anxiety for a score of 0–20, moderate anxiety for a score of 21–41, and severe anxiety for a score of 42–93.

2.2.5. Self-Evaluation Questionnaire (PSEQ)

The PSEQ is a scale developed by Lederman to assess women's adaptation to motherhood during the prenatal period [24]. Beydag and Mete conducted the validity and reliability evaluation of the scale in Turkish [25]. The scale has seven subdimensions, which are grouped as (a) thoughts about own well-being and well-being of the baby, (b) acceptance of pregnancy, (c) identification with motherhood role, (d) preparation for labor, (e) control in labor (fear of helplessness and loss of control in labor), (f) relationship with mother and (g) relationship with partner. Lower scores indicate higher levels of adaptation to pregnancy.

2.3. Statistical analysis

Descriptive statistics are presented as numbers, percentages, and median, minimum, and maximum values. The normality of distribution was determined with the Kolmogorov–Smirnov test. Chi-square and Fisher's exact tests were used to compare categorical variables, and the Mann–Whitney *U* test was used to compare continuous variables. The relationship between two continuous variables was evaluated using Spearman correlation analysis. Statistical significance was determined as $p < 0.05$. Correlation coefficients were evaluated as follows: 0.01–0.19=no or negligible relationship; 0.20–0.29=weak relationship; 0.30–0.39=moderate relationship; 0.40–0.69=strong relationship; 0.70–1=very strong relationship [26].

3. Results

3.1. Description of the study population

Initially, 78 pregnant adolescents and 155 pregnant adults were included in the study. A total of 50 first-trimester, 53 s-trimester, and 52 third-trimester individuals were evaluated for the study. Seventeen out of 78 pregnant adolescents and 39 out of 155 pregnant adults were excluded from the study due to incomplete questionnaires or study scales. Most of the pregnant adolescents resided in the rural areas of Hatay City, and they and their parents had a low level of education. A total of 177 pregnant women, consisting of 61 (34 %) pregnant adolescents and 116 (66 %) pregnant adults, were included in our study.

3.2. Characteristics of the pregnant adolescents

The minimum age of pregnant adolescents included in the study was 15 and the maximum was 19. None of the adolescents reported being victims of sexual abuse, and all had an early marriage (religious or state-recognized civil marriages), which was accepted and approved by the society in which they lived. The median age was 17.05 years old (16.23–18.74). Those under the age of 17 had religious marriages, and those over 17 were legally married. Four participants were under the age of 17, and 57 participants were over the age of 17 in the pregnant adolescent group. While the median number of years spent in educational settings was 8, all mothers ($n = 61$) and 94.6 % ($n = 56$) of the fathers were found to have a degree of secondary school grade or below. The spouses' age ranged from 19 to 47, with a median of 27. Seven pregnant adolescents (11.5 %) had experienced a pregnancy loss before, whereas 13 pregnant adolescents (21.3 %) had at least one living child. Nine out of 61 pregnant adolescents (14.8 %) reported having complications in their last pregnancy.

3.3. Comparison of the two groups (Table 1)

The level of education (8 [8–11] vs. 12 [11–14]; $p < 0.001$), level of maternal education (0% vs. 15 %; $p = 0.002$) and paternal education (5% vs. 25 %; $p = 0.002$), the employment rate (2% vs. 22 %; $p < 0.001$), and total monthly income (3500 [3000–5000] Turkish lira [TL] vs. 9000 [7000–12,000] TL; $p < 0.001$) were lower in the pregnant adolescents than in the pregnant adults. The presence of living children (57% vs. 21 %; $p < 0.001$) and the percentage of having a history of pregnancy loss (25% vs. 12 %; $p = 0.029$) were found to be higher in pregnant adults compared with pregnant adolescents. While the prevalence of pregnancy complications was 14.8 % in pregnant adolescents, this rate was found to be 6 % in pregnant adults ($p = 0.057$). The complications detected in the pregnant adolescent group were abortus imminens ($n = 3$), amniotic band syndrome ($n = 1$), oligohydramnios ($n = 2$), and bleeding ($n = 2$). The complications detected in the pregnant adult group were gestational diabetes mellitus (DM; $n = 3$), abortus imminens ($n = 2$), thrombocytopenia ($n = 1$), and thrombophilia ($n = 1$).

3.4. Description of the scale scores and level of psychopathology between the two groups

3.4.1. PSEQ scores

PSEQ–Thoughts about own well-being and the well-being of the baby were lower among the pregnant adolescents (18 [13–23]) than among the adults (21 [16–26]; $p = 0.007$). The PSEQ–Acceptance of pregnancy, the PSEQ–Identification with motherhood role, PSEQ–Relationship with mother, PSEQ–Relationship with partner, and PSEQ–Total scores were lower among pregnant adolescents (17 [16–21], 18 [17–22], 11 [10–13], 12 [10–14], 114 [101–134], respectively) than among the adult pregnant group (20 [17–24], 22 [19–25], 13 [11–18], 14 [11–16], and 129 [115–151], respectively; $p = 0.010$, $p < 0.001$, $p < 0.001$, $p = 0.006$, and $p < 0.001$, respectively) (Table 2).

Table 1
Sociodemographic and obstetric characteristics of participants.

		Adolescents <i>n</i> (%) [*]	Adults <i>n</i> (%) [*]	<i>p</i>
Age (year)		17.05 (16.23–18.74)	28.28 (25.59–31.89)	<0.001 [§]
Educational level (year)		8 (8–11)	12 (11–14)	<0.001 [§]
Employment status	Employed	1 (1.6)	24 (21.6)	<0.001 [‡]
	Unemployed	60 (98.4)	87 (78.4)	
Total monthly income (TL)		3500 (3000–5000)	9000 (7000–12,000)	<0.001 [§]
Maternal educational level	Middle school and below	61 (100.0)	62 (84.9)	0.002
	High school and above	0 (0.0)	11 (15.1)	
Paternal educational level	Middle school and below	56 (94.9)	55 (75.3)	0.002
	High school and above	3 (5.1)	18 (24.7)	
History of pregnancy loss	Absent	54 (88.5)	85 (74.6)	0.029 [‡]
	Present	7 (11.5)	29 (25.4)	
Presence of living child	Absent	48 (78.7)	50 (43.1)	<0.001 [‡]
	Present	13 (21.3)	66 (56.9)	
Planned pregnancy	Present	43 (70.5)	84 (73.7)	0.652 [‡]
	Absent	18 (29.5)	30 (26.3)	
Pregnancy stage	1st trimester	24 (39.3)	33 (28.4)	0.299 [‡]
	2nd trimester	16 (26.2)	40 (34.5)	
	3rd trimester	21 (34.4)	43 (37.1)	
Presence of complications in the last pregnancy [¶]	Present	9 (14.8)	7 (6.0)	0.057 [‡]
	Absent	52 (85.2)	109 (94.0)	

^{*} Column percentage is given. Continuous variables are shown with median (25th percentile–75th percentile) values.

[§] Mann–Whitney *U* test.

[‡] chi-square test.

[¶] Abortus imminens ($n = 6$), amniotic band syndrome ($n = 1$), thrombocytopenia ($n = 1$), gestational diabetes mellitus ($n = 3$), oligohydramnios ($n = 2$), thrombophilia ($n = 1$), bleeding ($n = 2$).

3.4.2. MSPSS subscale scores

MSPSS–Family and MSPSS–Special person scores were higher among pregnant adolescents (28 [26–28] and 28 [25–28], respectively) compared with pregnant adults [28 [24–28] and 24 [16–28], respectively; $p = 0.025$ and $p < 0.001$, respectively) (Table 2).

3.4.3. Depression and anxiety scores

BDI and PASS total scores were lower among adolescents (4 [1–10] and 15 [7–27], respectively) than among adults (8 [4–13] and 20 [14–35], respectively; $p = 0.001$ and $p = 0.021$, respectively) (Table 2). However, after grouping anxiety and depression levels according to the cut-off points, no statistical difference was observed regarding the level of anxiety and depression between the two groups (Table 3).

3.5. Relationship between social support and psychopathology (Table 4)

Social support scores were statistically higher in pregnant adolescents with minimal anxiety (81 [72–84]) than those with moderate–severe levels of anxiety (67 [59–77]; $p = 0.002$). In addition, adolescents with minimal–mild depression (79 [70–84]) had statistically higher social support scores than those with moderate–severe depression (61 [52–72]; $p = 0.002$). However, there was no statistically significant difference observed between social support scores and anxiety and depression levels in pregnant adults. When the presence of pregnancy complications and the PSEQ scores of the participants were compared, no statistical difference was observed in the adult group. Nevertheless, it was found that pregnancy complications in the

Table 2
Scale scores of pregnant adolescents and adults.

	Group						<i>p</i> [*]
	Adolescents			Adults			
	Median	25th Percentile	75th Percentile	Median	25th Percentile	75th Percentile	
PSEQ–Thoughts about own well-being and the well-being of the baby	18.00	13.00	23.00	21.00	16.00	26.00	0.007
PSEQ–Acceptance of pregnancy	17.00	16.00	21.00	20.00	17.00	24.00	0.010
PSEQ–Identification with motherhood role	18.00	17.00	22.00	22.00	19.00	25.00	<0.001
PSEQ–Preparation for labor	17.00	16.00	21.00	18.00	15.00	21.00	0.452
PSEQ–Control in labor	19.00	15.00	23.00	21.00	16.00	24.00	0.075
PSEQ–Relationship with mother	11.00	10.00	13.00	13.00	11.00	18.50	<0.001
PSEQ–Relationship with partner	12.00	10.00	14.00	14.00	11.00	16.50	0.006
PSEQ–Total	114.00	101.00	134.00	129.50	115.50	151.00	<0.001
PASS score	15.00	7.00	27.00	20.00	14.00	35.50	0.021
BDI score	4.00	1.00	10.00	8.00	4.00	13.00	0.001
MSPSS–Family	28.00	26.00	28.00	28.00	24.00	28.00	0.025
MSPSS–Friend	25.00	16.00	28.00	25.00	19.00	28.00	0.725
MSPSS–Special person	28.00	25.00	28.00	24.00	16.00	28.00	<0.001
MSPSS–Total	77.00	65.00	84.00	72.00	60.00	83.00	0.155

PSEQ: Prenatal Self-Evaluation Questionnaire, MSPSS: Multidimensional Scale of Perceived Social Support.

BDI: Beck Depression Inventory, PASS: Perinatal Anxiety Screening Scale.

* Mann–Whitney *U* test.

Table 3
Comparison of the level of anxiety and depression in pregnant adolescents and adults.

		Adolescents		Adults		<i>p</i> *
		<i>n</i>	%	<i>n</i>	%	
Anxiety	Minimal level	37	60.7	59	50.9	0.447
	Moderate level	17	27.9	42	36.2	
	Severe level	7	11.5	15	12.9	
Depression	Minimal level	45	73.8	71	61.2	0.216
	Mild level	7	11.5	27	23.3	
	Moderate	7	11.5	16	13.8	
	Severe Level	2	3.3	2	1.7	

* Chi-square test.

Table 4
The relationship between the MSPSS scores of the participants and the level of anxiety and depression in pregnant adolescents and adults.

			MSPSS score			<i>p</i> -value*
			Median	25. p	75. p	
Anxiety	Adolescent	Minimal level	81	72	84	0.002
		Moderate to severe levels	67	59	77	
	Adult	Minimal level	75	61	83	0.338
		Moderate to severe levels	70	53	83	
Depression	Adolescent	Minimal to mild levels	79	70	84	0.002
		Moderate to severe levels	61	52	72	
	Adult	Minimal to mild levels	74	60	83	0.065
		Moderate to severe levels	64	42	77	

MSPSS: Multidimensional Scale of Perceived Social Support, 25. p: 25. Percentile, 75. p: 75. Percentile.

* Mann Whitney *U* test.

adolescent group decreased adaptation to pregnancy in the *Acceptance of pregnancy* and *Relationship with partner* subdimensions ($p = 0.022$ and $p = 0.046$, respectively). No correlation was found between the PSEQ *Relationship with partner* subdimension and BDI scores and between the PSEQ *Relationship with mother* subdimension and PASS scores in the adult group. A weak and moderate correlation existed between BDI

scores or PASS scores with other PSEQ subdimensions. On the other hand, in the pregnant adolescent group, the scores of all PSEQ subdimensions and the total score showed moderate and strong correlations with BDI and PASS scores in general. Furthermore, the PSEQ *control in labor* score in the adolescent group was positively and strongly correlated with BDI scores ($r = 0.638$, $p < 0.001$). In addition, a very strong positive correlation was found between PSEQ *thoughts about own well-being and the well-being of the baby*, PSEQ *control in labor*, and PASS scores ($r = 0.722$, $p < 0.001$) in the adolescent group.

4. Discussion

This study examined the well-being of pregnant adolescents by assessing the presence of pregnancy complications, the impact of these complications on women’s perceptions of pregnancy and acceptance of it, and the protective role of social support. The findings of the study indicate that pregnancy complications tend to be more common in pregnant adolescents than in pregnant adults (adolescents: 14.8 %, adults: 6.1 %, $p = 0.057$), and the presence of complications adversely affects acceptance of pregnancy, especially in the adolescent group. In addition to supporting previous studies reporting more common pregnancy complications in adolescents [7], this study shows that complications negatively affect the acceptance of pregnancy in adolescents compared with adults in a statistically significant way. Biological immaturity is considered the primary cause of poor birth outcomes for teenagers, who are still developing [27]. In addition, the adverse effects of complications that may occur in connection with these unfavorable outcomes on acceptance of pregnancy are also probably related to the adolescents’ immature cognitive, emotional, and social status. Based on these reasons, it appears that acceptance of pregnancy is more adversely affected in adolescents as compared with pregnant adults.

A significant relationship was found in this study between pregnancy complications and poor partner relationships in pregnant adolescents. Marriage can bring about notable changes in a teenage girl’s life, requiring her to suddenly transition into adult roles and responsibilities before she is developmentally ready to handle such obligations [28]. Becoming a mother without proper preparation for the challenges of teenage motherhood and attempting to manage new roles and responsibilities while still dealing with unresolved developmental tasks can pose significant difficulties [29]. Research indicates that social support has a myriad of positive impacts on pregnant adolescents. It facilitates the improvement of coping mechanisms for the challenges of pregnancy [30], aids in the transition to maternity [31], and serves as a buffer against the negative consequences of anxiety [32,33].

Adolescents are more prone than adults to experience strained relationships with their partners due to pregnancy, pregnancy complications, and the responsibility they must assume within marriage before attaining adequate maturity. In our research, pregnancy among adolescent and adult women occurred as a result of “marriage,” whether it be a religious or state-recognized civil marriage. Thus, regardless of whether one lives in an extended or nuclear family, marriage brings responsibilities such as cooking and cleaning the house. Since adolescents have not yet reached sufficient cognitive maturity, their ability to cope with the problems they experience is probably not as adequate as that of adults, which may negatively affect their relationships with their spouses. As a result, in pregnant adolescents, who are a group with a high risk of pregnancy complications, the adverse effects of pregnancy complications on both the acceptance of pregnancy and the relationship with the spouse might reduce social support and adversely affect the attachment to be formed with the child. Considering that the attachment between mother and child begins during the prenatal period, poor acceptance of pregnancy is likely to have a negative impact on the attachment between the mother and the child [34]. In the literature, it is reported that various sociodemographic and obstetric factors may affect prenatal attachment negatively by disrupting adaptation to pregnancy in high-risk pregnancies [35].

Studies of pregnant adolescents have shown that adolescent pregnancy is closely associated with low economic status [36]; lower income levels and education are reported to lead to higher adolescent pregnancy rates [37]. In line with previous research, this study determined that the educational levels of the participants and their parents, as well as household income levels, were statistically significantly lower in pregnant adolescents than in pregnant adults. Consistent with our findings, a previous study also reported that women who gave birth to their first child before the age of 20 were more likely to have grown up in lower socioeconomic conditions than women who postponed childbearing until adulthood [2].

A study conducted in South Africa reported that adolescent mothers had a higher risk of dropping out of school [38]. In our study, one out of five pregnant adolescents claimed they had to drop out of school to get married. Adolescence plays a critical role in providing time for a young woman to develop her social, cognitive, and professional skills before assuming reproductive responsibilities [39]. Pregnancy in adolescence hinders this process, disrupting adolescent girls' education and causing them to need more social support as they have not yet developed these skills [40].

Although the protective role of social support against depression during pregnancy is well known in the literature [10,12], there are fewer studies on anxiety. One study found that pregnant adolescents with an anxiety disorder reported lower levels of perceived social support [33]. In one third of women with postpartum depression, depressive symptoms begin during pregnancy, and if untreated, about 20 % of cases progress to chronic depression [41]. Social support is known to decrease the likelihood of experiencing depression during pregnancy, leading to favorable health and pregnancy outcomes [42]. Our study found that perceived social support led to a statistically significant decrease in depression and anxiety scores in pregnant adolescents during the prenatal period compared with adults. The findings of another study indicate that inadequate social support plays a crucial role in adverse maternal and newborn outcomes in adolescent pregnancies, independent of early maternal age, which supports the significant role of social support in adolescent pregnancy [5].

Our research findings establish that the relationship status of pregnant adolescents with their mothers or spouses has a stronger correlation with their anxiety and depression scores in comparison with pregnant adults. These results indicate that pregnant teenagers need support from their social circle more, and their levels of anxiety and depression tend to rise as their relationship quality deteriorates. Studies have shown that adolescent mothers who receive support from their spouses experience reduced psychological distress and depression as

well as increased life satisfaction and self-esteem [43]. The effect of the relationship with a spouse during pregnancy on depression, as seen in our study, supports the findings of this previous study.

4.1. Limitations and strengths of the study

One limitation of the study was its cross-sectional design, which prevented capturing the causal directions of associations between the factors we investigated. Hence, there is a requirement for future prospective studies. Another limitation was the use of scales instead of semi-structured interviews to evaluate psychiatric disorders, as subjective evaluation may have been a disadvantage in determining frequency. The adolescent group did not have as many participants as initially calculated, with only half of the participants compared with the pregnant adult group; thus, generalizing the findings is challenging due to the limited number of participants in the adolescent group. Despite these limitations, the study possesses several strengths. For instance, pregnant adults were included in the study for comparison with pregnant adolescents, contributing to the robustness of our findings. Additionally, both depression and anxiety symptoms were evaluated in this study; lastly, the relationship between the presence of pregnancy complications and mental illness was assessed.

5. Conclusion

The most significant contribution to the literature made by this study is its emphasis on the negative consequences of getting married at a young age, even in traditional societies where child marriage is accepted. Considering the various disadvantages outlined in the study, it is imperative to conduct more extensive research focusing on environmental, social, economic, and familial factors associated with adolescent pregnancy, thus enabling the development of intervention programs and policies to prevent these pregnancies in the future. In order to develop effective social policies aimed at preventing adolescent pregnancies, it is imperative to conduct further studies that contribute to our understanding of the underlying cultural and social processes. In addition, future studies should consider variables that may attenuate the aggravating impact of sociocultural factors on adolescent pregnancy and assess modifiable social factors that may elucidate the association between adolescent pregnancy and long-term mental health outcomes.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declaration of competing interest

The authors have no relevant financial or nonfinancial interests to disclose.

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